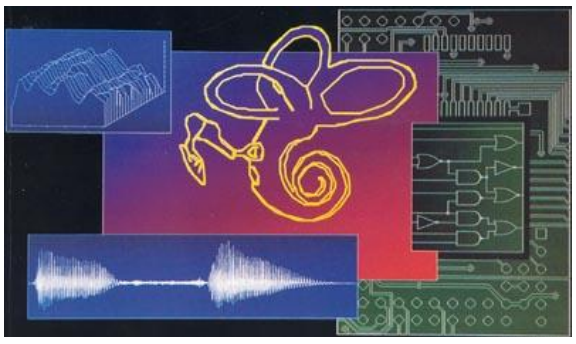
Improving cochlear implants by modelling how auditory neurons „dance to the beat“



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1 Introduction

Cochlear implant is a neuroprosthetic device for cases of severe to profound hearing loss and is used today by over 300.000 patients, especially by children and babies who are yet to learn to communicate („pre-lingual deafness“).

The technolgy behind CI has been around since ´60ies, but the main principle has not changed in the meantime: a cochlear implant bypasses the damaged mechanical part of the inner ear and directly stimulates the auditory nerves using electrodes [1], thus producing a hearing sensation.

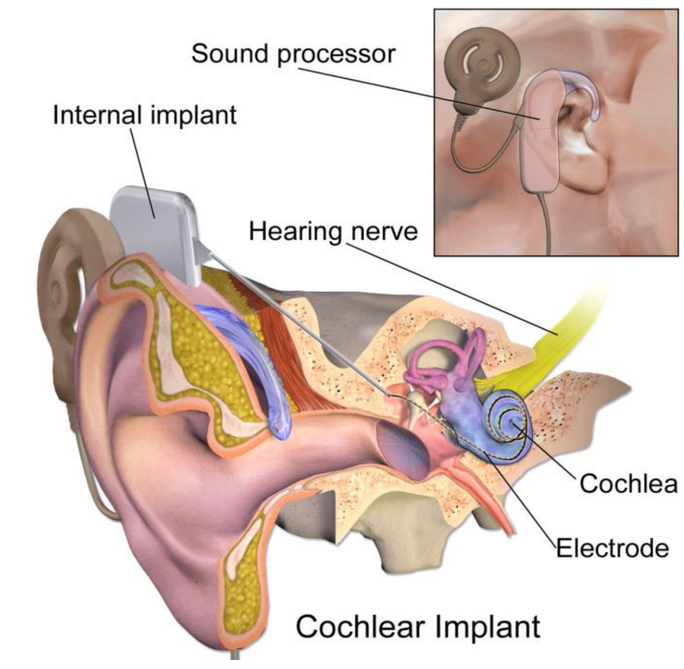

Fig. 1. Schematic of human ear with a cochlear implant.
Source: Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2)

Typical stimulation strategy for CI is to [2]:
1) Separate measured sound into n frequency bands
2) Extract the **envelope** (energy vs. time) for each frequency band
3) Generate electrical stimulation proportional to the energy for each frequency band (or m<n bands), and
4) Send the signal to the corresponding electrodes (*i.e.* positions along the basilar membrane)

2 Motivation

CI performance is **optimized for understanding speech** – for which only the sound envelope is important – while the fine temporal structure of the sound, necessary to appreciate **music** or understand **tonal languages** (*e.g.* Mandarin), is not the main concern. Similarly, **pitch perception** and intelligibility of **speech in noise** („cocktail party“ scenario) are limited, reducing the overall quality of sound percieved by CI patients, and thus limiting their ability to fully rehabilitate into the „hearing“ society.

In order to alleviate these issues new coding strategies need to be developed, and the basic knowledge necessary to reach this goal and compare different strategies is knowing **how do the auditory neurons respond to electrical stimulation** – or conversely, how to ellicit a desired temporal activity pattern in the auditory nerves.

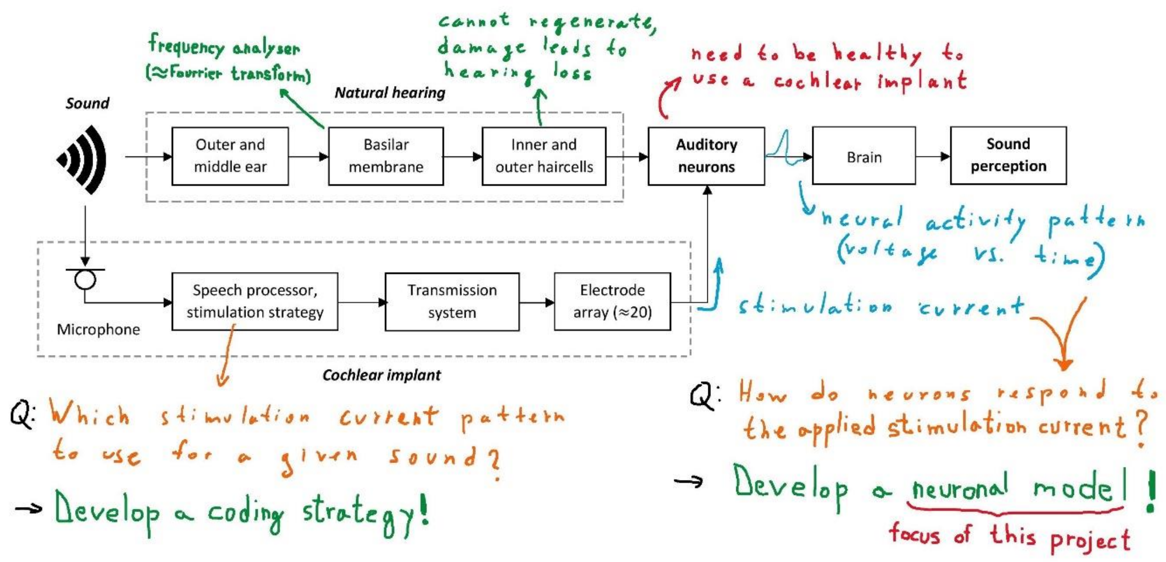

Fig. 2. Overview of normal hearing process and cochlear implant hearing process. Adapted from Hess [3].

3 Neural Model

To allow for real-time computations needed for CIs, while still retaining important features of neuronal behaviour, a phenomenological integrate-and-fire model is to be used. In this model, the neuronal membrane is modelled as a **capacitor with finite resistance** driven by an external **stimulation current** – and when the neuron´s voltage crosses a **threshold**, a **spike** is generated and the voltage is then reset to a lower value.

As an extension to this simple model, four prominent features of neural behaviour are to be added (as factors influencing the dynamic firing threshold): refractoriness, spike rate adaptation, facilitation, and adaptation.
Additionally, the inherent **stohasticity** of neuronal firing needs to be added as noise in the membrane potential, **spatial spread** of stimulus current is to be taken into account, and an **ensemble** of neurons is to be simulated (*ca.* 10.000 neurons).

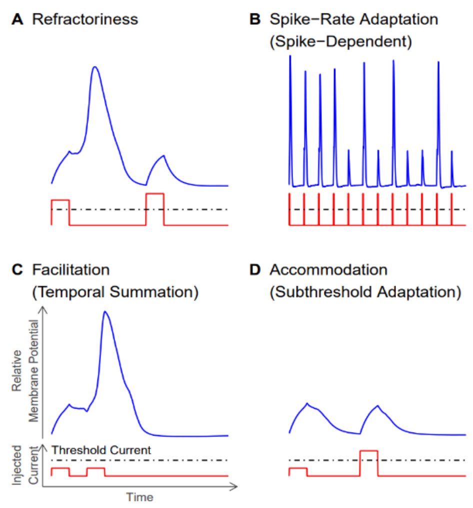

Fig. 3. Four characteristic behaviours of neurons included in the model, caused by ion channel (de)activation or passive membrane charging (facilitation). Stimulation current is shown in red, and corresponding neuron membrane voltage is shown in blue. Source: Boulet [4]

4 Expected benefits and further work

Given that most of the current models **underpredict** the adaptation observed in experimental data since they do not include both the active adaptation and spike rate adaptation, it is expected that our model will produce more accurate predicitions (as has been already shown in study by Boulet *et al.* [4]).
It is expected that our model will allow for development of novel stimulation strategies, leading to better hearing for CI users, and on the research side, „allow for easier investigation of mechanisms influencing CI stimulation“ [4].

Further work includes, among other, the following:
1) Comparison of model predictions with ECAP measurements for various input signals (*e.g.* modulated pulse trains, important for speech) and refinement of the model
2) Development and testing of a coding strategy based on the obtained model
3) Development of a measurement protocol for experimentally determining the model parameters for each subject

5 References

1) Thomas Haslwanter *et al.*; Sensory Systems, Wikibook, accessed 16.04.2018.
2) Charlène Batrel, Benjamin Chaix, Sam Irving; *Promenade around the cochlea / Rehabilitation / Signal processing*, NeurOreille, accessed 15.04.2018.
3) Max Hess; *Modelling of electrically evoked compound action potentials (ECAPs) in response to amplitude-modulated pulse trains*, ETHZ semester project, 2017
4) Jason Boulet, Sonia Tabibi, Norbert Dillier, Ian C. Bruce; *Phenomenological Model of Auditory Nerve Fiber Responces to Cochlear Implant Stimulation: Subtreshold and Suprathreshold Influences*, paper under review